Experiment 4 – Train Reservation System

Aim:

To develop an online Train Reservation System for searching trains, checking seat availability, booking tickets, generating PNR numbers, checking PNR status, and managing cancellations and refunds.

Algorithm:

1. Log in using valid passenger credentials.
2. Enter journey date, passenger details, origin, destination, and class.
3. Display available trains and seat availability.
4. Select the required train and make payment.
5. Generate the ticket and PNR number after successful payment.
6. Store payment, train, fare, and reservation details.
7. Allow passengers to check PNR status.
8. Allow ticket cancellation and calculate the refundable amount.
9. Allow admin to maintain train, fare, PNR, and reservation information.

Code:

```
<!DOCTYPE html>
<html>
<head>
<title>Train Reservation System</title>

<style>
*{box-sizing:border-box}
```

body{font-family:Arial;margin:0;background:#f3f4f6}

header{background:#174ea6;color:white;text-align:center;padding:20px}

nav{background:#222;padding:12px;text-align:center}

nav button{background:none;color:white;border:0;margin:5px;cursor:pointer}

.box{background:white;width:90%;max-width:1000px;margin:20px auto;padding:20px;border-radius:10px}

input,select{padding:10px;margin:5px;width:95%}

button{padding:10px 15px;background:#174ea6;color:white;border:0;border-radius:5px;cursor:pointer}

button:hover{background:#0d3575}

.hidden{display:none}

.train{border:1px solid #ccc;padding:15px;margin:10px 0;border-radius:8px}

.seat{display:inline-block;padding:12px;margin:5px;background:#ddd;cursor:pointer}

.selected{background:green;color:white}

.booked{background:red;color:white;cursor:not-allowed}

.success{color:green;font-weight:bold}

.error{color:red;font-weight:bold}

table{width:100%;border-collapse:collapse}

td,th{border:1px solid #aaa;padding:8px;text-align:center}

</style>

</head>

<body>

```
<header>
<h1>🚆 Train Reservation System</h1>
<p>Search • Reserve • Check PNR • Cancel</p>
</header>

<nav>
<button onclick="show('home')">Home</button>
<button onclick="show('register')">Register</button>
<button onclick="show('login')">Login</button>
<button onclick="show('search')">Search Train</button>
<button onclick="show('pnr')">PNR Status</button>
<button onclick="show('history')">My Bookings</button>
<button onclick="show('admin')">Admin</button>
<button onclick="logout()">Logout</button>
</nav>

<!-- HOME -->

<div class="box" id="home">
<h2>Welcome</h2>
<p>Online Train Reservation System</p>
<ul>
<li>Train search</li>
<li>Seat availability</li>
<li>Online reservation</li>
```

PNR generation

PNR status

Ticket cancellation

Refund calculation

Booking history

Admin management

<!-- REGISTER -->

<h2>Passenger Registration</h2>

<input id="rname" placeholder="Full Name">

<input id="remail" placeholder="Email">

<input id="rphone" placeholder="Phone">

<input id="rpass" type="password" placeholder="Password">

<button onclick="register()">Register</button>

<!-- LOGIN -->

```
<div class="box hidden" id="login">
<h2>Passenger Login</h2>

<input id="lemail" placeholder="Email">
<input id="lpass" type="password" placeholder="Password">

<button onclick="login()">Login</button>

<p id="loginmsg"></p>
</div>

<!-- SEARCH -->

<div class="box hidden" id="search">

<h2>Search Train</h2>

<input id="from" placeholder="Origin Station">
<input id="to" placeholder="Destination Station">
<input id="date" type="date">

<select id="classType">
<option>Sleeper</option>
<option>AC 3 Tier</option>
<option>AC 2 Tier</option>
<option>First Class</option>
```

```
</select>

<button onclick="searchTrain()">Search</button>

<div id="trainList"></div>

</div>

<!-- BOOKING -->

<div class="box hidden" id="booking">

<h2>Reserve Ticket</h2>

<p id="selectedTrain"></p>

<h3>Select Seat</h3>

<div id="seats"></div>

<h3>Passenger Details</h3>

<input id="pname" placeholder="Passenger Name">
<input id="page" type="number" placeholder="Age">

<select id="pgender">
```

<option>Male</option>

<option>Female</option>

<option>Other</option>

</select>

<button onclick="confirmBooking()">Pay & Book Ticket</button>

<!-- PNR -->

<h2>Check PNR Status</h2>

<input id="pnrInput" placeholder="Enter PNR Number">

<button onclick="checkPNR()">Check PNR</button>

```
<!-- HISTORY -->

<div class="box hidden" id="history">

<h2>My Booking History</h2>

<div id="historyData"></div>

</div>

<!-- ADMIN -->

<div class="box hidden" id="admin">

<h2>Admin Panel</h2>

<h3>Add Train</h3>

<input id="aname" placeholder="Train Name">
<input id="anum" placeholder="Train Number">
<input id="afrom" placeholder="From">
<input id="ato" placeholder="To">
<input id="aprice" type="number" placeholder="Fare">
<input id="aseats" type="number" placeholder="Seats">
```

```
<button onclick="addTrain()">Add Train</button>

<h3>Train List</h3>

<div id="adminTrains"></div>

<h3>All Bookings</h3>

<div id="adminBookings"></div>

</div>

<script>

let currentUser=localStorage.getItem("currentUser");
let selectedTrain=null;
let selectedSeat=null;

/* DEFAULT TRAINS */

if(!localStorage.getItem("trains")){

let trains=[
{
id:1,
name:"Chennai Express",
```

```
number:"12601",
from:"Chennai",
to:"Bangalore",
price:500,
seats:20
},
{
id:2,
name:"Coimbatore Express",
number:"12673",
from:"Chennai",
to:"Coimbatore",
price:600,
seats:15
},
{
id:3,
name:"Kerala Express",
number:"12625",
from:"Chennai",
to:"Kochi",
price:800,
seats:25
}
];
```

```
localStorage.setItem("trains",JSON.stringify(trains));

}

/* SHOW PAGE */

function show(id){

document.querySelectorAll(".box").forEach(x=>{
x.classList.add("hidden");
});

document.getElementById(id).classList.remove("hidden");

if(id=="history")loadHistory();

if(id=="admin")loadAdmin();

}

/* REGISTER */

function register(){

let users=JSON.parse(localStorage.getItem("users")||"[]");
```

```
let user={
name:rname.value,
email:remail.value,
phone:rphone.value,
password:rpass.value
};

if(!user.name||!user.email||!user.password){

regmsg.innerHTML="<span class='error'>Fill all details</span>";
return;

}

if(users.some(x=>x.email==user.email)){

regmsg.innerHTML="<span class='error'>Email already
registered</span>";
return;

}

users.push(user);

localStorage.setItem("users",JSON.stringify(users));
```

```
regmsg.innerHTML="<span class='success'>Registration
successful</span>";

}

/* LOGIN */

function login(){

let users=JSON.parse(localStorage.getItem("users")||"[]");

let user=users.find(x=>
x.email==lemail.value &&
x.password==lpass.value
);

if(user){

localStorage.setItem("currentUser",user.email);
currentUser=user.email;

loginmsg.innerHTML="<span class='success'>Login successful</span>";

}else{

loginmsg.innerHTML="<span class='error'>Invalid email or
password</span>";
```

```
}

}

/* LOGOUT */

function logout(){

localStorage.removeItem("currentUser");
currentUser=null;

alert("Logged out");

show("home");

}

/* SEARCH TRAIN */

function searchTrain(){

let fromValue=from.value.toLowerCase();
let toValue=to.value.toLowerCase();

let trains=JSON.parse(localStorage.getItem("trains"));
```

```
let result=trains.filter(x=>
x.from.toLowerCase().includes(fromValue) &&
x.to.toLowerCase().includes(toValue)
);

trainList.innerHTML="";

if(result.length==0){

trainList.innerHTML="<p>No trains found</p>";
return;

}

result.forEach(t=>{

trainList.innerHTML+=`

<div class="train">

<h3>${t.name}</h3>

<p>Train No: ${t.number}</p>
<p>Route: ${t.from} → ${t.to}</p>
<p>Fare: ₹${t.price}</p>
```

```
<p>Available Seats: ${t.seats}</p>

<button onclick="openBooking(${t.id})">
Reserve Ticket
</button>

</div>

`;

});

}

/* OPEN BOOKING */

function openBooking(id){

if(!currentUser){

alert("Please login first");
show("login");
return;

}
```

```
let trains=JSON.parse(localStorage.getItem("trains"));

selectedTrain=trains.find(x=>x.id==id);

selectedSeat=null;

selectedTrainText();

createSeats();

show("booking");

}

/* TRAIN DETAILS */

function selectedTrainText(){

selectedTrain.innerHTML="";

}

/* CREATE SEATS */

function createSeats(){
```

```
let html="";

for(let i=1;i<=selectedTrain.seats;i++){

html+=`
<span class="seat"
onclick="selectSeat(this,${i})">
${i}
</span>
`;

}

seats.innerHTML=html;

}

/* SELECT SEAT */

function selectSeat(element,num){

document.querySelectorAll(".seat").forEach(x=>{
x.classList.remove("selected");
});

element.classList.add("selected");
```

```
selectedSeat=num;

}

/* BOOK */

function confirmBooking(){

if(!currentUser){

alert("Login required");
return;

}

if(!selectedSeat){

alert("Select a seat");
return;

}

if(!pname.value||!page.value){

alert("Enter passenger details");
```

```
return;

}

let bookings=
JSON.parse(localStorage.getItem("bookings")||"[]");

let pnr="PNR"+Math.floor(100000+Math.random()*900000);

let booking={

pnr:pnr,

email:currentUser,

train:selectedTrain.name,

number:selectedTrain.number,

from:selectedTrain.from,

to:selectedTrain.to,

seat:selectedSeat,

passenger:pname.value,
```

```
age:page.value,

classType:classType.value,

fare:selectedTrain.price,

status:"CONFIRMED"

};

bookings.push(booking);

localStorage.setItem("bookings",JSON.stringify(bookings));

/* REDUCE SEAT */

let trains=
JSON.parse(localStorage.getItem("trains"));

let t=trains.find(x=>x.id==selectedTrain.id);

t.seats--;

localStorage.setItem("trains",JSON.stringify(trains));
```

```
bookmsg.innerHTML=`

<p class="success">

Ticket booked successfully!<br><br>

<strong>PNR: ${pnr}</strong><br>

Train: ${booking.train}<br>

Seat: ${booking.seat}<br>

Fare: ₹${booking.fare}

</p>

`;

}

/* PNR STATUS */

function checkPNR(){

let bookings=
JSON.parse(localStorage.getItem("bookings")||"[]");
```

```
let b=bookings.find(x=>x.pnr==pnrInput.value);

if(!b){

pnrResult.innerHTML=
"<p class='error'>PNR not found</p>";

return;

}

pnrResult.innerHTML=`

<h3>PNR Details</h3>

<p>PNR: ${b.pnr}</p>
<p>Passenger: ${b.passenger}</p>
<p>Train: ${b.train}</p>
<p>Route: ${b.from} → ${b.to}</p>
<p>Seat: ${b.seat}</p>
<p>Class: ${b.classType}</p>
<p>Fare: ₹${b.fare}</p>
<p>Status: <b>${b.status}</b></p>

${b.status=="CONFIRMED" ?
```

```
`<button onclick="cancelTicket('${b.pnr}')">
Cancel Ticket
</button>`:""}

`;

}

/* HISTORY */

function loadHistory(){

if(!currentUser){

historyData.innerHTML="Please login first";
return;

}

let bookings=
JSON.parse(localStorage.getItem("bookings")||"[]");

let mine=bookings.filter(x=>x.email==currentUser);

if(mine.length==0){
```

```
historyData.innerHTML="No bookings found";
return;

}

let html="<table>";

html+=`
<tr>
<th>PNR</th>
<th>Train</th>
<th>Seat</th>
<th>Fare</th>
<th>Status</th>
<th>Action</th>
</tr>
`;

mine.forEach(b=>{

html+=`

<tr>

<td>${b.pnr}</td>
```

```
<td>${b.train}</td>
<td>${b.seat}</td>
<td>₹${b.fare}</td>
<td>${b.status}</td>

<td>

${b.status=="CONFIRMED"?

`<button onclick="cancelTicket('${b.pnr}')">
Cancel
</button>`:"Cancelled"}

</td>

</tr>

`;

});

html+="</table>";

historyData.innerHTML=html;

}
```

```
/* CANCEL */

function cancelTicket(pnr){

let bookings=
JSON.parse(localStorage.getItem("bookings")||"[]");

let b=bookings.find(x=>x.pnr==pnr);

if(!b)return;

let refund=b.fare*0.75;

b.status="CANCELLED";
b.refund=refund;

localStorage.setItem("bookings",JSON.stringify(bookings));

alert(
"Ticket cancelled\nRefund Amount: ₹"+refund
);

checkPNR();

loadHistory();
```

```
}

/* ADMIN ADD TRAIN */

function addTrain(){

let trains=
JSON.parse(localStorage.getItem("trains"));

let train={

id:Date.now(),

name:aname.value,

number:anum.value,

from:afrom.value,

to:ato.value,

price:Number(aprice.value),

seats:Number(aseats.value)
```

```
};

trains.push(train);

localStorage.setItem("trains",JSON.stringify(trains));

alert("Train added");

loadAdmin();

}

/* ADMIN */

function loadAdmin(){

let trains=
JSON.parse(localStorage.getItem("trains"));

adminTrains.innerHTML="";

trains.forEach(t=>{

adminTrains.innerHTML+=`

<div class="train">
```

```
<b>${t.name}</b><br>

Train No: ${t.number}<br>

${t.from} → ${t.to}<br>

Fare: ₹${t.price}<br>

Seats: ${t.seats}

</div>

`;

});

/* BOOKINGS */

let bookings=
JSON.parse(localStorage.getItem("bookings")||"[]");

adminBookings.innerHTML="";

bookings.forEach(b=>{
```

```
adminBookings.innerHTML+=`

<div class="train">

PNR: ${b.pnr}<br>
Passenger: ${b.passenger}<br>
Train: ${b.train}<br>
Seat: ${b.seat}<br>
Fare: ₹${b.fare}<br>
Status: ${b.status}

</div>

`;

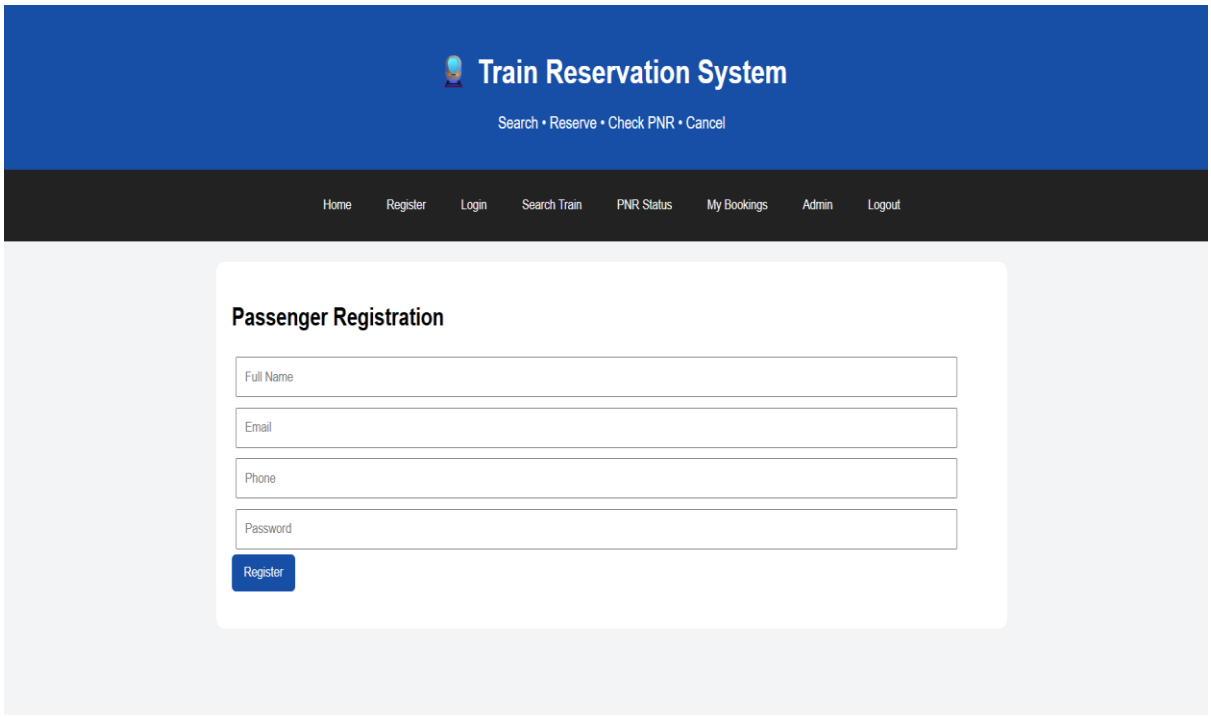
});

}

</script>

</body>
</html>
```

Output:



Result:

The Train Reservation System was successfully developed with train search, seat availability, ticket booking, PNR generation, PNR status checking, cancellation, and refund management.